



NUTRITIONAL ASSESSMENT (HuNu3081)

CHAPTER SIX

NUTRITIONAL ASSESSMENT OF HOSPITAL PATIENTS

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Objectives

At the end of this chapter you will be able to:

- Discuss why nutritional assessment of Hospital patients is needed.
- Define what nutrition screening mean?
- Identify the purpose of nutrition assessment of hospital patients.
- Identify criteria for a nutritional screening tools.
- Identify the circumstances when nutrition screening should be undertaken.
- Identify the nutritional screening tools (indices).

Introduction

- Though more than 30% of all patients in hospitals are undernourished, nutritional risk continues to be unrecognized and undertreated.
- Prevention of malnourishment with **early intervention** is much **more effective** in improving outcomes than reacting once a patient has become ill & has obvious nutritional deficits.
- Malnutrition in hospital patients is associated with:
 - ✓ Impaired immunity to infection, greater morbidity, mortality, delayed recovery, prolonged hospital stay & higher healthcare costs.
 - So, maintaining or improving the nutritional status of patients admitted to hospitals will **improve clinical outcomes** and **help to reduce health care costs**.

6.1. Nutritional Assessment of Patients

- In clinical medicine nutritional assessment was introduced during the 1970s, when protein-energy malnutrition (PEM) was first recognized among hospital patients in North America.
- PEM is most prevalent in hospital patients, notably among the **elderly** even today, but less than the 1970s.
 - So, most of the tests in hospitals assess PEM.
- Tests may include dietary, anthropometric, static biochemical, and functional indices of nutritional status.

Nutritional Assessment of Patients...(1)

- Nutritional assessment identifies patients at nutritional risk early & helps to prevent the development of a state of nutritional depletion or excess.
 - For example, a patient with cancer who is undergoing a major surgical procedure is more likely to develop wound healing problems and/or infections if he/she has lost a significant amount of weight prior to surgery.
- Patient's nutritional state affects the entire course of illness and response to therapeutic interventions.
- Interventions for malnutrition now range from **diet modification** and **counseling** to specialized nutrition therapies such as **enteral** or **parenteral nutrition**.

Nutritional Assessment of Patients... (2)

- Nutritional screening and assessment are best performed by a **multidisciplinary team** (including; nutritionist/dietitians, nurses, physicians, pharmacist etc).
- Numerous protocols have been developed for screening patients for nutritional risk, as well as for conducting full nutritional assessment.
- The screening protocols used should be:
 - ✓ Simple, inexpensive, and be completed by existing staff,
 - ✓ Have the ability to anticipate nutritional depletion, so as to prevent before malnutrition becomes clinically significant.

Nutritional Assessment of Patients... (3)

- In many hospitals, nutrition screening is routinely applied to **high risk patients**, (those who are **at risk of becoming malnourished** or **who are malnourished**).
- So that, appropriate nutritional support can be implemented to improve the nutritional status of patients at risk.
- This practice is important because malnutrition among both medical and surgical patients is an important determinant of post-operative mortality and morbidity.

Nutritional Assessment of Patients...(4)

- Systemic illness causes dysmetabolism (alteration in the metabolism of energy substrates).
- Catabolic illness leads to decrement in serum proteins, and also a reduction in the lean mass.
- More than 70% of hospital patients (over 70yrs) had hypoalbuminemia.
- Undernourished patients do not become hypoalbuminemic until and unless they become ill.
- On the other hand, ill patients may become hypoalbuminemic without being malnourished.

Nutritional Assessment of Patients...(5)

- **Kwashiorkor** is the state of dysmetabolism seen in ill patients and frequently associated with starvation.
- Occur due to the misuse of protein by the body as an energy substrate instead of as a building block.
- It occurs in children who have developed Hypoalbuminemia and ascites (accumulation of fluid in the peritoneal cavity resulted from low level of albumin).
- All patients are at nutritional risk, once catabolism and hypoproteinemia have developed so, close monitoring of intake by nutrition specialists should be considered.

Table 6.1. Factors affecting serum albumin, prealbumin & Transferrin levels

Factors	Effects
Inflammatory states (infections, catabolic stress, post operative states, burns, cancer etc.), hepatic failure, protein losing state (nephrotic syndrome & enteropathies), use of corticosteroids, eclampsia, over hydration, kwashiorkor etc.	Decreases concentrations of serum albumin
Dehydration	Increases serum albumin
Advanced renal disease and use of oral contraceptives	Increases prealbumin and transferrin
Iron deficiency No-iron deficiency anemia & use of corticosteroids	Increases Transferrin Decreases Transferrin

6.2. Malnutrition and Body weight

- **Recent weight loss** appears to be the most important single indicator of nutritional status among patients.
- Most guidelines identify patients at nutritional risk if they:
 - ✓ Are less than 80% of ideal weight,
 - ✓ Have a body mass index less than 20kg/m^2 ,
 - ✓ Have lost 1-2% in one week, 5% of baseline or 5 pounds in one month, (**significant weight loss**).
 - ✓ Have lost 7.5% in 3 months, 10 pounds or 10% of usual body weight in 6 months (**significant weight loss**).

6.3. Assessing Nutritional Risk

- Measuring a patient's weight at every visit, or having the patients weigh themselves at home is the simplest & most effective method for monitoring nutritional status.
- During the physical examination, notice if:
 - ✚ Consistent weight loss, and the presence of rashes.
 - ✚ Zinc deficiency: can occur in patients with chronic diarrhea and causes rash on the shins.
 - ✚ Lanugo hair: a late sign of severe malnourishment.
 - ✚ Presence of muscle wasting: put the patient at very high nutritional risk.

Lanugo hair



6.4. Purpose of Nutrition Assessment of Hospital patients

The purposes of nutritional assessment of patients include:

- ☑ To determine a patient's nutritional status,
 - ☑ To identify clinically relevant malnutrition (nutrient & metabolic risks) and nutrition-related problems, their causes & significance.
 - ☑ To establish a nutrition care plan & to monitor changes in the patient's nutritional status.
- The assessment often includes patient history, medical diagnosis and treatment plan, nutrition and medication histories, nutrition-related physical examination, psychological, social, and environmental aspects.

6.5. Malnutrition Screening

- Malnutrition screening enables the early identification of patients who may benefit from nutrition support.
- Routine nutrition screening of hospitalized patients is recommend.
- Malnutrition can be successfully treated through basic nutrition support initiatives implemented by healthcare professionals, or by in-depth dietary plans produced by dietitians.
 - However, this cannot be achieved unless the individuals are first identified through nutrition screening.

Malnutrition Screening...(1)

- Malnutrition can affect a person's physical & psychological health by:
 - ✦ Impairing the immune function leading to an increased risk of infections (mobility) and sepsis.
 - ✦ Delaying wound healing and increasing the risk of ulcers.
 - ✦ Causing muscle wasting and weakness leading to a deterioration in respiratory function and an increased risk of chest infections, impaired cardiac function/ cardiac failure.
 - ✦ Causing apathy and depression leading to loss of morale & reduced recovery.
 - ✦ Causing a general sense of weakness leading to a reduced appetite, exacerbating the current situation.

6.5.1.What is Nutrition Screening?

- The process of identifying individuals who are malnourished or who are at risk of malnutrition and determining if a detailed nutrition assessment is indicated (American Society for Enteral & Parenteral Nutrition).
- It is a powerful way of identifying malnourished individuals in need of nutritional support.
- Nutrition screening should be used as a precursor to a nutritional assessment.
 - So, it should be completed during the first contact with a patient.

Nutrition Screening cont'd...

- Any nutrition screening tool should be **easy to use, quick, economical, standardized, and validated**.
- Screening tools should be both sensitive & specific, and if possible, predictors of the success of nutritional therapy.
- Information is collected on the patient's change in weight, BMI or weight history, adequacy of food intake and severity of disease.

Why Nutrition Screening?

- To rapidly identify patients who are at high nutritional risk or have poor nutritional status at hospital admission.

6.6. What Makes A Good Nutrition Screening Tool?

- There are 71 identified nutritional screening/ assessment tools used in clinical practices.
- A screening tool should be able to identify;
 - ✦ Chronic protein-energy status: such as BMI.
 - ✦ Recent changes in protein-energy status: such as weight loss or reduction in dietary intake.
 - ✦ The likelihood of any future changes in protein-energy status, *i.e.* any loss of weight or reduction in dietary intake.

Table 6.2. Criteria for selecting nutritional screening tools

Practical	➤ It is quick and easy to complete and to understand ➤ It has a range of alternative measures for when weight or height cannot be measured
Universal	➤ It can be used in all adults, of all ages (including the elderly), including the sick and healthy ➤ It is applicable across different care settings (hospital, GP practice, nursing home, community) ➤ It allows continuity of care ➤ It can be used for public health purposes
Reliable	➤ It provides good reproducibility between users ➤ It has good internal reliability
Valid	➤ It has face (extent to which the tool appears to measure what it is supposed to measure), content, internal, concurrent and predictive validity
Evidence-based and peer-reviewed	
Linked to a care plan for treatment	➤ It facilitates nursing and other staff to initiate appropriate monitoring or treatment and referral to the dietitian or nutrition support team
Developed by a multidisciplinary group or used by all healthcare professionals	
Acceptable to patients and healthcare professionals	

6.7. When Nutrition Screening Should be Undertaken?

- Nutrition screening should be undertaken:
 - ☑ On admission: (all hospital inpatients on admission and all outpatients at their first clinic appointment should be screened).
 - ☑ On initial registration at general practice surgeries,
 - ☑ When there is a clinical concern for outpatients; and
 - ☑ It should be repeated on a regular basis (weekly) for long-stay patients.

6.8. Screening using Single Index

- Biochemical, anthropometric measurements & sometimes functional tests are used for screening based on a single index.
- Abnormal values for more than one parameter frequently occur in hospital patients, (*i.e.* no single index is completely satisfactory with regard to sensitivity and specificity).
- When single indices are individually used, their clinical importance & predictive value is uncertain.
- Predictive value is the best measure of clinical usefulness of any index of nutritional status.
 - It measures the ability of an index to correctly predict the presence or absence of nutrition associated complications; and other health or nutrition outcomes.

6.8. Screening using Single Index ...Cont'd

- Some of the single indices are relatively **insensitive**.
 - For example: serum albumin & transferrin has relatively long half-lives, and hence they respond slowly to nutritional depletion and repletion.
- Their sensitivity & specificity may also be affected by non-nutritional factors; including: hepatic diseases, nephrosis, and etc.
- Indeed there is no consensus as to which index can best be applied to determine nutritional status or risk of malnutrition.

Parameters used as a single nutritional assessment indices

Anthropometric measurements

- Weight (k.g), height (cm): BMI
- Usual weight (k.g),
- Ideal body weight (IBW) (k.g)
- Weight as percentage of IBW (%)
- Weight as percentage of usual weight (%)
- Triceps skinfold (mm)
- MUAC/ Arm muscle circumference(cm)

Diet and Nutritional status

- Protein intake (g), Caloric intake (kJ)
- Nitrogen balance (g)
- Obligatory nitrogen loss (g)
- Net protein utilization
- Basal energy expenditure (BEE) (kJ/day)
- Caloric intake as % of BEE (%)
- Skin test result (mm)

Biochemical tests

Serum albumin (g/L)

Total iron binding capacity (µg/dl)

Serum transferrin (mg/dl)

Lymphocyte count (%)

WBC count (no./mm³)

Total Lymphocyte count (no./mm³)

24-hr urinary nitrogen (g)

24-hr urinary creatinine (mg)

Creatinine height index as % of standard (%)

6.8.2. Multi-parameter Screening

- Dealt with multiple components: medical history, dietary intake, amount of weight loss, biochemical variables, and anthropometric measurements.
- Multi-parameter indices of nutritional status used to increase sensitivity, specificity & predictive value.
- To be effective, these multi-parameter indices (“scoring systems”) must be easy to use, cost effective, and tested for reproducibility, validity, sensitivity & specificity.
- They include:
 - ✓ Subjective Global Assessment (SGA),
 - ✓ Nutritional Risk Index (NRI),
 - ✓ Mini Nutritional Assessment (MNA),
 - ✓ Nutritional Risk Screening Tool 2002 (NRS-2002),
 - ✓ Malnutrition Universal Screening Tool (MUST).

6.8.2.1. Subjective Global Assessment (SGA)

- Useful to detect **surgical patients** with **established malnutrition**.
- It is based on carefully performed medical history & physical examination, and no anthropometric or laboratory indices are included.
- The basis of SGA is to determine whether nutrient assimilation has been restricted because of:
 - Decreased food intake, mal-digestion, or malabsorption, any effects of malnutrition on organ function & body composition, and whether the patient's disease process influences nutrient requirements.
- It needs specialized training.

Subjective Global Assessment (SGA)... cont'd

- SGA is performed by a trained physician on five areas of medical history.
 - Body weight lost in the previous 6 months: mild (5%), moderate (5–10%), & severe (10%).
 - Changes in dietary intake: normal or abnormal.
 - Presence of persistent gastrointestinal symptoms (anorexia (loss of appetite), nausea (urge to vomit), vomiting, diarrhea, & abdominal pain etc. occurs daily for 2 weeks).
 - The patient's functional capacity or energy level: bedridden, sub-optimally active, or full capacity.
 - The metabolic demands of the patient's underlying disease state, *e.g.* high-stress illnesses (burns, major trauma, & severe inflammation such as acute colitis).

6.8.2.2. Nutritional Risk Index (NRI)

- Uses serum albumin, and present weight compared with usual weight, so can capture both poor clinical outcome and nutritional risk.
- Designed for **surgical patients**.

$$\text{NRI} = (1.519 \times \text{serum albumin (g/L)}) + 41.7 \times (\text{present weight/usual weight})$$

NRI > 100: no malnutrition;

NR I= 97.5-100: mild malnutrition,

NR = 83.5-97.5: moderate malnutrition;

NRI < 83.5: severe malnutrition.

- **Usual weight:** the stable weight above 6 months before admission.

6.8.2.3. Mini Nutritional Assessment (MNA)

- Developed specifically for detecting malnutrition in **elderly hospital patients**, because of the poor performance of many other multi-parameter indices in this age group.
- Helps to detect patients who need preventive nutritional measures.
- The original form used for MNA consists of:
 - Anthropometric assessment (weight, height, & weight loss).
 - General assessment (life style, medication, morbidity).
 - Dietary assessment (number of meals, food & fluid intake, and autonomy of feeding).
 - Subjective assessment (self perception of health & nutrition).
- It could be used by relatively untrained personnel, and took less than 20 minutes to complete.

6.8.2.4. Nutritional Risk Screening Tool 2002 (NRS-2002)

- Used to detect the presence of undernutrition and/or its risk of development among hospitalized patients and identify patients who require more in-depth nutritional assessment and periodic monitoring.

Nutritional Risk Screening (NRS 2002)

Table 1 Initial screening

		Yes	No
1	Is BMI <20.5?		
2	Has the patient lost weight within the last 3 months?		
3	Has the patient had a reduced dietary intake in the last week?		
4	Is the patient severely ill ? (e.g. in intensive therapy)		

Yes: If the answer is 'Yes' to any question, the screening in Table 2 is performed.

No: If the answer is 'No' to all questions, the patient is re-screened at weekly intervals. If the patient e.g. is scheduled for a major operation, a preventive nutritional care plan is considered to avoid the associated risk status.

NRS-2002 ...Cont'd

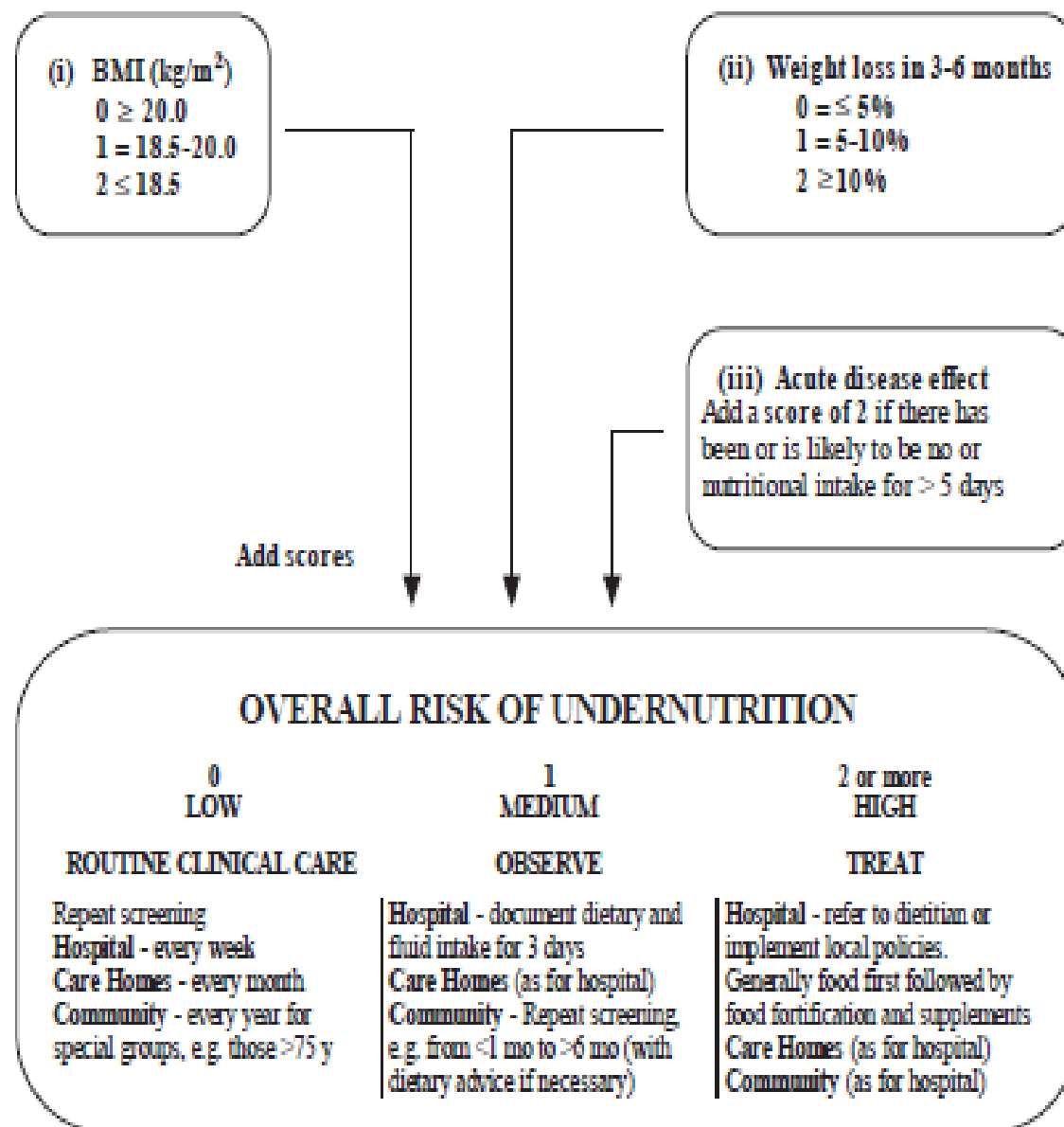
Table 2 Final screening

Impaired nutritional status		Severity of disease (≈ increase in requirements)	
Absent Score 0	Normal nutritional status	Absent Score 0	Normal nutritional requirements
Mild Score 1	Wt loss > 5% in 3 mths or Food intake below 50–75% of normal requirement in preceding week	Mild Score 1	Hip fracture* Chronic patients, in particular with acute complications: cirrhosis*, COPD*. <i>Chronic hemodialysis, diabetes, oncology</i>
Moderate Score 2	Wt loss > 5% in 2 mths or BMI 18.5 – 20.5 + impaired general condition or Food intake 25–60% of normal requirement in preceding week	Moderate Score 2	Major abdominal surgery* Stroke* <i>Severe pneumonia, hematologic malignancy</i>
Severe Score 3	Wt loss > 5% in 1 mth (> 15% in 3 mths) or BMI < 18.5 + impaired general condition or Food intake 0–25% of normal requirement in preceding week in preceding week.	Severe Score 3	Head injury* Bone marrow transplantation* <i>Intensive care patients (APACHE > 10).</i>
Score:	+	Score:	= Total score
Age	if ≥ 70 years: add 1 to total score above = age-adjusted total score		
Score ≥ 3: the patient is nutritionally at-risk and a nutritional care plan is initiated Score < 3: weekly rescreening of the patient. If the patient e.g. is scheduled for a major operation, a preventive nutritional care plan is considered to avoid the associated risk status.			

6.8.2.5. Malnutrition Universal Screening Tool (MUST)

- MUST is appropriate & evidence-based nutrition screening tool, that meets all of the screening tool criteria (listed in table 6.2).
- It has a five-step processes using a scoring system to identify adults at risk of malnutrition.
- **Step 1:** calculate BMI by measuring height & weight (score 0–2).
- **Step 2:** assesses weight loss score by measuring unplanned weight loss over the past 3–6 months (score 0–2).
- **Step 3:** gives an acute disease effect score of 2 if the patient is acutely ill, or unlikely to be no nutritional intake for >5 days.
- **Step 4:** involves adding the scores from steps 1–3 to give a total risk score for malnutrition.
- **Step 5:** involves putting into place a treatment plan according to the total score achieved.

Malnutrition Universal Screening Tool (MUST) for adults



Can be adapted for special circumstances (e.g. when weight and height cannot be measured or when there are fluid disturbances) using specified alternative measurements including subjective criteria. It also identifies obesity (BMI > 30 kg/m²).

Key points

- The purpose of nutritional screening is to rapidly identify patients at high nutritional risk.
- The purpose of nutritional assessment, however, is to define a patient's nutritional status, to define clinically relevant malnutrition and to monitor changes in nutritional status.
- Nutritional risk continues to be unrecognized and undertreated in clinical practice.
- Hospitals need to develop comprehensive strategies for identifying and treating malnutrition at hospital admission and for monitoring the nutritional status of patients during their hospital stay.

End of the course.

THANK YOU!